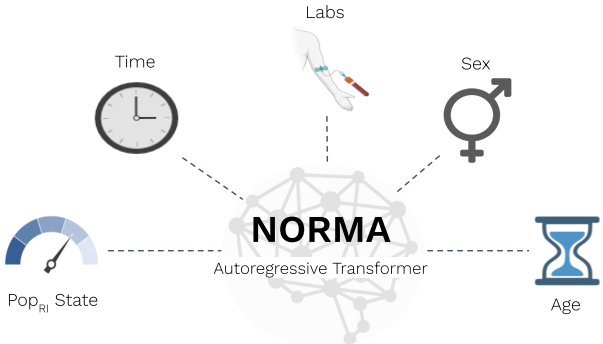
A Contextual Sequence Modeling



Predicts distribution for next "healthy" lab

B Training and Validation Cohorts

TRAINING		VALIDATION	
EHRSHOT	**MIMIC-IV**	**CLALIT**	**eICU**
4.6k patients 91k sequences	114k patients 1.8M sequences	1M patients 16M sequences	98k patients 91k sequences

20 years | **4M** sequences | **30** labs

Complete Blood Count | Hepatic Function | Metabolic Function | Lipid Panel

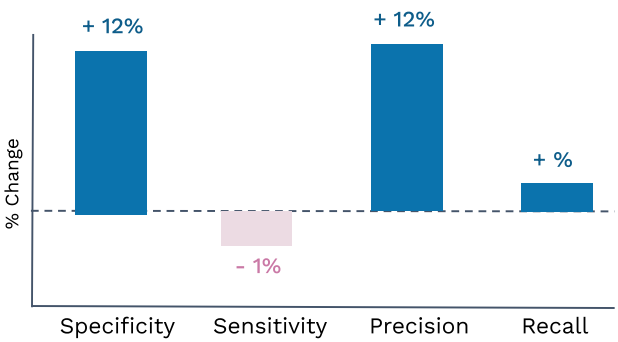
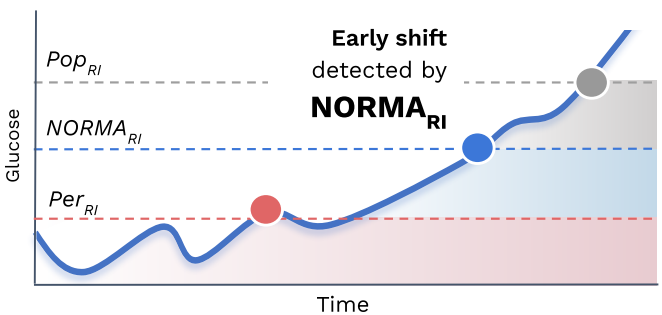
C Reference Interval Paradigms

Pop$_{RI}$	Per$_{RI}$	NORMA$_{RI}$
LDL (mg/mL)	LDL (mg/mL)	LDL (mg/mL)
Population	**Individualized**	**Contextualized**
✗ Personalized	✗ Robust to noise	✓ Personalized
✗ Repeat Labs	✗ Time-Sensitive	✓ Population-Aware
	✗ Handles sparsity	✓ Time-Sensitive
		✓ Models Uncertainty

D Clinical Utility of NORMA$_{RI}$

Glucose → Type 2 Diabetes

Pop$_{RI}$	Per$_{RI}$	NORMA$_{RI}$
Normal	*High*	*High*



E Early Risk Stratification